

Bilinear Forms 1

Arham Lodha

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Problem 1a. For $V = F^n$, let A be a square matrix. Then define a form by $\langle v, w \rangle_A := v^T A w$, for column vectors v, w . Show all bilinear forms on V occur this way.

Proof. $\forall v, w, z \in V, \forall a, b \in F, \langle av + bw, z \rangle_A = (av + bw)^T A z = av^T A z + bw^T A z = a \langle v, z \rangle_A + b \langle w, z \rangle_A$.
 $\langle v, aw + bz \rangle_A = v^T A (aw + bz) = v^T A a w + v^T A b z = a \langle v, w \rangle_A + b \langle v, z \rangle_A$. The form is bilinear.

Let $k = \dim V$. Let $V_B = v_1, \dots, v_k$ be a basis of V , such that $v = \sum_{i=1}^k a_i v_i = A V_B$ and $w = \sum_{i=1}^k b_i v_i = B V_B$. Let $\langle v, w \rangle$ be some bilinear form. Define a matrix $M \ni M_{i,j} = \langle v_i, v_j \rangle$.
 $\langle v, w \rangle = \langle \sum_{i=1}^k a_i v_i, \sum_{j=1}^k b_j v_j \rangle = \sum_{i,j} a_i b_j \langle v_i, v_j \rangle = \sum_{i,j} a_i m_{i,j} b_j = A^T M B$. $\square$

Problem 1b. Thus once you pick a basis for finite $\dim V$, a bilinear form is the same thing as a matrix. We call the form symmetric if $\langle v, w \rangle = \langle w, v \rangle$ check this is the same thing as A being symmetric.

Proof. Suppose $\langle v, w \rangle = \langle w, v \rangle$, a symmetric bilinear form for all $v, w \in V$. $v^T A w = w^T A v = (\langle w, v \rangle)^T = v^T A^T w \implies A = A^T$.

Suppose $A = A^T \implies v^T A w = v^T A^T w = (v^T A^T w)^T = w^T A v \implies \langle v, w \rangle = \langle w, v \rangle$. $\square$

Problem 1c. Show that when we change basis, the matrix A changes not (in general) by conjugation, but rather by $A \rightarrow P^t A P$, for invertible P .

Proof. Let B_1 be the tuple of the original basis. Let B_2 be the tuple of the new basis. Let P be the matrix which changes basis from B_2 to B_1 . Let X_1 and Y_2 be coordinate vectors. Let $v = X_1 = P X_2$ and $w = Y_1 = P Y_2$. $\langle v, w \rangle_A = v^T A w = X_1^T A Y_1 = (P X_2)^T A (P Y_2) = X_2^T P^T A P Y_2$. So under the new basis the new matrix is $P^T A P$. $\square$

Problem 2. In 1) we see that once we choose a basis, a bilinear form is "the same thing" as a square matrix. But a matrix is also "the same thing" as a linear map, and a bilinear form $\langle, \rangle$ on V is definitely not the same thing as a linear map $V \rightarrow V$. Instead show $v \rightarrow \langle v, \rangle$ gives a linear map $V \rightarrow V^* := \text{Hom}_F(V, F)$. Show that a bilinear form IS the same thing as a linear map $V \rightarrow V^*$.

Proof. Let $\phi : V \rightarrow V^* \ni v \rightarrow \langle v, \rangle$. Let $a, b \in F, v, w \in V, \phi(av + bw) = \langle av + bw, \rangle = a \langle v, \rangle + b \langle w, \rangle = a \phi(v) + b \phi(w)$.

Let S be the set of bilinear forms on V . Let $H := \text{Hom}_F(V, V^*)$. Let $\alpha : S \rightarrow H \ni \langle, \rangle \rightarrow [v \rightarrow \langle v, \rangle]$. Let $\beta : H \rightarrow S$ such that $\forall h \in H$ create a bilinear form $\langle, \rangle \ni \forall v, w \in V, \langle v, w \rangle = h(v)(w)$ send h to this map.

$\forall v, w \in V, (\beta \circ \alpha)(\langle, \rangle)[v, w] = \beta([v \rightarrow \langle v, \rangle])[v, w] = \langle v, w \rangle$.

$\forall v, w \in V, (\alpha \circ \beta)(h)(v)(w) = \alpha([x \rightarrow h(x)(y)])(v)(w) = h(v)(w)$. Thus α is a bijection.

$\forall v, w \in V, \forall p, q \in B, \forall f, g \in F, \alpha(fp + gq)(v) = [x \rightarrow (fp + gq)(x,)](v)(w) = (fp + gq)(v, w) = fp(v, w) + gq(v, w) = f\alpha(p)(v)(w) + g\alpha(q)(v)(w)$

Thus α is a linear map. Thus $S \cong H$ as vector spaces. □

Problem 3. One of the main results of elementary linear algebra (a proof of which we give below) is that a real symmetric matrix is orthogonally diagonalizable. If $A = A^T$ there exists orthogonal P with $PAP^{-1} = \text{diagonal}$. Note $P^{-1} = P^t$, so P^tAP is diagonal. Lets prove a much more general version of this last statement:

Theorem: Let V be finite dim over F , with $\text{char}(F) \neq 2$, and let $\langle, \rangle$ be a symmetric bilinear form on V . Then there is a basis $b_1, \dots, b_n$ with $\langle b_i, b_j \rangle = 0$ for $i \neq j$.

Show the theorem is equivalent to the statement that if A is a square symmetric matrix over V , there is an invertible P such that $P^tAP = \text{diagonal}$.

Solution. Create a symmetric bilinear form $b \ni \forall v, w \in V, b(v, w) = v^T Aw$, since A is symmetric. By the theorem, $\exists V_B = \{b_1, \dots, b_n\}$ a basis such that $b(b_i, b_j) = 0$ for $i \neq j$. Let P be the matrix from V_B to the original basis. Let $v = Pv'$ and $w = Pw'$. So $b(v, w) = b(Pv', Pw') = v'^T P^T APw'$. Since under this basis, $b(b_i, b_j) = 0 = b_i^T P^T AP b_j \implies P^T AP$ is diagonal. ■

Problem 4a. Here we prove the above theorem. For this we consider the associated "quadratic form" $Q(v) := \langle v, v \rangle$. Obviously $\langle, \rangle$ determines Q . Show, for $\text{char} F \neq 2$, that the other direction holds, Q determines $\langle, \rangle$. In fact show $\langle v, w \rangle = -\frac{1}{2}(Q(v) + Q(w) - Q(v + w))$.

Proof. $\forall v, w \in V, -\frac{1}{2}(\langle v, v \rangle + \langle w, w \rangle - \langle v + w, v + w \rangle) = -\frac{1}{2}(\langle v, v \rangle - \langle v, v + w \rangle + \langle w, w \rangle - \langle w, v + w \rangle) = -\frac{1}{2}(\langle v, v \rangle - \langle v, v \rangle - \langle v, w \rangle + \langle w, w \rangle - \langle w, v \rangle - \langle w, w \rangle) = \frac{1}{2}(\langle v, w \rangle + \langle w, v \rangle) = \langle v, w \rangle$. So $\langle v, w \rangle = \frac{1}{2}(Q(v + w) - Q(v) - Q(w))$. □

Claim 4b. Show the natural addition map $\text{span}(v) \oplus v^\perp \rightarrow V$ is an iso (where $\oplus$ means direct sum) where $Q(v) \neq 0$ and $v^\perp = \{w \mid \langle v, w \rangle = 0\}$.

Proof. $\forall k \in F/\{0\}, \langle kv, v \rangle = kQ(v) \neq 0 \implies kv \notin v^\perp \implies \forall w \in \text{span}(v), w \notin v^\perp$.

Let $A : \text{span}(v) \oplus v^\perp \rightarrow V$ be the natural addition iso.

Because $v^\perp$ is a subspace, since $v \notin v^\perp \implies v$ linearly independent from all $w \in v^\perp$. Thus the kernel of the natural addition map is 0 and A is injective.

Surjectivity: Let $\phi : V \rightarrow F \ni x \rightarrow \langle v, x \rangle$. Thus $v^\perp = \ker(\phi)$. By rank nullity, $\dim V = \dim(\phi_*(V)) + \dim \ker \phi \implies \dim \ker \phi = \dim V - 1$. Since $\dim \text{span}(v) = 1$ and $\text{span}(v) \cap v^\perp = 0 \implies \dim \text{span}(v) + \dim v^\perp = \dim V \implies A$ is Surjective.

Thus A is a bijection. $\forall (v_1, w_1), (v_2, w_2) \in \text{span}(v) \oplus v^\perp$ and $\forall a, b \in F, A(a(v_1, w_1) + b(v_2, w_2)) = A(av_1 + bv_2, aw_1 + bw_2) = av_1 + aw_1 + bv_2 + bw_2 = aA(v_1, w_1) + bA(v_2, w_2)$. Thus A is a linear map. Thus A is an iso. □

Claim 4c. Now to prove the theorem. We proceed by induction on dimension, when V has dim 1 there is nothing to prove. If $\langle, \rangle$ is identically zero, of course we are done. If not, show there is a vector with $\langle v, v \rangle \neq 0$. consider $v^\perp := \{w, \langle v, w \rangle = 0\}$. Now consider $v^\perp$ and proceed by induction.

Proof. If $\langle, \rangle = 0$, we are done.

Suppose $\exists v, w \in V \ni \langle v, w \rangle \neq 0$.

$(k = 2)$: Let V be a vector space with dimension 2. Suppose $\forall v \in V, Q(v) = 0 \implies \forall w \in V, \langle v, w \rangle = 0$ (by formula above), which breaks the assumption. So exists $v \in V \ni Q(v) \neq 0$. $\forall w \in v^\perp/\{0\}$, by

definition, $\langle v, w \rangle = 0$. Additionally as proven above v and w are linearly independent so v and w are such a basis.

Inductive Hypothesis: Suppose for all vector spaces V where $\dim V = k$, exists a basis $b_1, \dots, b_k \ni \langle b_i, b_j \rangle = 0$ for $i \neq j$.

Suppose V is a vector space of dimension $k+1$. Let $n = k+1$. Suppose $\forall v \in V, Q(v) = 0 \implies \forall w \in V, \langle v, w \rangle = 0$ (by formula above), which breaks the assumption. So exists $v \in V \ni Q(v) = 0$. Since $A : \text{span}(v) \oplus v^\perp \rightarrow V$ is a iso, take the basis of $v^\perp, b_1, \dots, b_k$. By definition, $\langle v, b_i \rangle = 0$ and $\langle b_i, b_j \rangle = 0$. Additionally as proven above, $av + \sum_{i=1}^k c_i b_i = 0 \iff a, c_i = 0$ because v is linearly independent from $v^\perp$. Thus $v, b_1, \dots, b_k$ is a basis where the bilinear form applied to any pair equals 0.

□